

Belinda Z. Li

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Education

Massachusetts Institute of Technology

2020 –

PhD candidate, Electrical Engineering and Computer Science

- **Masters degree** awarded Feb. 2023

- Advisor: Jacob Andreas

University of Washington

2015 – 2019

Bachelors of Science, Computer Science, Magna Cum Laude

- Advisor: Luke Zettlemoyer

Publications

Peer-Reviewed Publications

- [2] **Belinda Z. Li**, Zifan Carl Guo, and Jacob Andreas. “(How) Do Language Models Track State?” In: *Forty-second International Conference on Machine Learning*. 2025. URL: <https://openreview.net/forum?id=8SXosAVIFH>.
- [3] **Belinda Z. Li**, Been Kim, and Zi Wang. “QuestBench: Can LLMs ask the right question to acquire information in reasoning tasks?” In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. 2025. URL: <https://openreview.net/forum?id=gpwA9aZLTZ>.
- [4] **Belinda Z. Li**, Alex Tamkin, Noah Goodman, and Jacob Andreas. “Eliciting Human Preferences with Language Models”. In: *The Thirteenth International Conference on Learning Representations*. 2025. URL: <https://openreview.net/forum?id=LvDwwAgMEW>.
- [5] **Belinda Z. Li**, Emmy Liu, Alexis Ross, Abbas Zeitoun, Graham Neubig, and Jacob Andreas. “Language Modeling with Editable External Knowledge”. In: *Findings of 2025 Annual Conference of the Nations of the Americas Chapter of the ACL*. 2025. URL: <https://openreview.net/forum?id=x5pPbgRR0t>.
- [6] Andi Peng, **Belinda Z. Li**, Ilia Sucholutsky, Nishanth Kumar, Julie Shah, Jacob Andreas, and Andreea Bobu. “Adaptive Language-Guided Abstraction from Contrastive Explanations”. In: *8th Annual Conference on Robot Learning*. 2024. URL: <https://openreview.net/forum?id=OGjGtN6hoo>.
- [8] Andi Peng, Andreea Bobu, **Belinda Z. Li**, Theodore Summers, Ilia Sucholutsky, Nishanth Kumar, Thomas Griffiths, and Julie Shah. “Preference-Conditioned Language-Guided Abstractions”. In: *Proceedings of the ACM/IEEE International Conference on Human Robot Interaction (HRI 2024)*. ACM/IEEE. 2024.
- [9] Andi Peng, Ilia Sucholutsky, **Belinda Z. Li**, Theodore Summers, Thomas Griffiths, Jacob Andreas, and Julie Shah. “Learning with language-guided state abstractions”. In: *RSS Workshop on Social Intelligence in Humans and Robots*. 2023.
- [10] **Belinda Z. Li**, Jason Eisner, Adam Pauls, and Sam Thomson. “[Toward Interactive Dictation](#)”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 15319–15338.
- [11] Evan Hernandez, **Belinda Z. Li**, and Jacob Andreas. “Inspecting and Editing Knowledge Representations in Language Models”. In: *First Conference on Language Modeling*. 2024. URL: <https://openreview.net/forum?id=AdtL6fgNRv>.
- [12] **Belinda Z. Li**, William Chen, Pratyusha Sharma, and Jacob Andreas. “LaMPP: Language Models as Probabilistic Priors for Perception and Action”. In: *ICLR 2024 Workshop on Generative Models for Decision Making*. under review. 2024. URL: <https://openreview.net/forum?id=1aMfd925b5>.
- [13] **Belinda Z. Li**, Maxwell Nye, and Jacob Andreas. “[Language Modeling with Latent Situations](#)”. In: *Findings of the Association for Computational Linguistics: ACL 2023*. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 12556–12571.

- [14] **Belinda Z. Li**, Jane Yu, Madian Khabsa, Luke Zettlemoyer, Alon Halevy, and Jacob Andreas. “[Quantifying Adaptability in Pre-trained Language Models with 500 Tasks](#)”. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Seattle, United States: Association for Computational Linguistics, July 2022, pp. 4696–4715.
- [15] Qinyuan Ye, **Belinda Z. Li**, Sinong Wang, Benjamin Bolte, Hao Ma, Wen-tau Yih, Xiang Ren, and Madian Khabsa. “[On the Influence of Masking Policies in Intermediate Pre-training](#)”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 7190–7202.
- [16] **Belinda Z. Li**, Maxwell Nye, and Jacob Andreas. “[Implicit Representations of Meaning in Neural Language Models](#)”. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Online: Association for Computational Linguistics, Aug. 2021, pp. 1813–1827.
- [17] Nayeon Lee, **Belinda Z. Li**, Sinong Wang, Pascale Fung, Hao Ma, Wen-tau Yih, and Madian Khabsa. “[On Unifying Misinformation Detection](#)”. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, June 2021, pp. 5479–5485.
- [18] **Belinda Z. Li**, Sewon Min, Srinivasan Iyer, Yashar Mehdad, and Wen-tau Yih. “[Efficient One-Pass End-to-End Entity Linking for Questions](#)”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Online: Association for Computational Linguistics, Nov. 2020, pp. 6433–6441.
- [19] Nayeon Lee, **Belinda Z. Li**, Sinong Wang, Wen-tau Yih, Hao Ma, and Madian Khabsa. “[Language Models as Fact Checkers?](#)” In: *Proceedings of the Third Workshop on Fact Extraction and VERification (FEVER)*. Association for Computational Linguistics, July 2020, pp. 36–41.
- [20] **Belinda Z. Li**, Gabriel Stanovsky, and Luke Zettlemoyer. “[Active Learning for Coreference Resolution using Discrete Annotation](#)”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, July 2020, pp. 8320–8331.

Preprints

- [1] **Belinda Z. Li**, Zifan Carl Guo, Vincent Huang, Jacob Steinhardt, and Jacob Andreas. *Training Language Models to Explain Their Own Computations*. 2025.
- [7] Kunal Handa, Yarin Gal, Ellie Pavlick, Noah Goodman, Jacob Andreas, Alex Tamkin, and **Belinda Z. Li**. *Bayesian Preference Elicitation with Language Models*. 2024.
- [21] Sinong Wang, **Belinda Z. Li**, Madian Khabsa, Han Fang, and Hao Ma. *Linformer: Self-Attention with Linear Complexity*. 2020.
- [22] Qinyuan Ye, **Belinda Z. Li**, Sinong Wang, Benjamin Bolte, Hao Ma, Xiang Ren, Wen-tau Yih, and Madian Khabsa. *Studying Strategically: Learning to Mask for Closed-book QA*. 2020.

Industry Experience

Transluce AI <i>Research Fellow</i>	May 2025 –
Letta AI <i>Research Fellow</i>	May 2025 –
Google Deepmind <i>Student Researcher</i>	June 2024 – Jan. 2025
Microsoft Semantic Machines <i>Research Intern</i>	June 2022 – Aug. 2022
Facebook AI	June 2021 – Aug. 2021

Research Intern, AI Integrity Team

Facebook AI

Aug. 2019 – Sep. 2020

Software Engineer, AI Integrity Team (Facebook AI Applied Research)

- Built multi-lingual, multi-modal (image + text) representations for Facebook posts. Shipped to hate speech domain.

<https://ai.facebook.com/blog/ai-advances-to-better-detect-hate-speech/>

Facebook

June 2018 – Aug. 2018

Software Engineer Intern, Bot Detection/Traffic Quality Team

- Worked on automated bot detection using mouse movements.

Awards

Rising Stars in EECS	2024
Clare Boothe Luce Graduate Fellowship	2021
National Defence Science and Engineering Graduate Fellowship	2021
National Science Foundation Graduate Research Fellowship (<i>Declined</i>)	2021
Ida M. Green Memorial Fellowship	2020 – 2021
Denice Dee Denton Scholars Endowment	2018 – 2019
UW Dean's List	2015 – 2017 (annual), SP18/F18/SP19 (quarterly) (every eligible quarter)
National Merit Scholar	2016

Talks

Invited Talks

TTIC Young Researcher Seminar Series	January 2026
Bacconi University (MilaNLP), <i>Training Language Models to Explain Their Own Computations</i>	January 2026
Northeastern University (Bau Lab), <i>Training Language Models to Explain Their Own Computations</i>	January 2026
Executive Code Podcast, <i>QuestBench: Can LLMs Ask the Right Questions?</i>	July 2025
Plutous AI Fireside Chat, <i>(How) Do Language Models Track State?</i>	April 2025
Toronto Data Workshop, <i>Eliciting Tasks and Preferences with Language Models</i>	May 2024
MILA, <i>World State in Language Models</i>	October 2023
ServiceNow, <i>World State in Language Models</i> ,	October 2023
NAACL Conference Oral, <i>Quantifying Adaptability in Pre-trained Language Models with 500 Tasks</i>	July 2022
Google N2Formal Team, <i>Implicit Representations of Meaning in Neural Language Models</i>	September 2021
MIT EI Workshop, <i>Implicit Representations of Meaning in Neural Language Models</i>	September 2021
Google DeepMind, <i>Implicit Representations of Meaning in Neural Language Models</i>	July 2021
Primer AI, <i>Implicit Representations of Meaning in Neural Language Models</i>	June 2021

Contributed Talks

USC Allegro Lab, <i>(How) Do Language Models Track State?</i>	August 2025
Stanford Digital Economy Lab, <i>Can Language Models Ask Useful Questions?</i>	July 2025
Berkeley NLP, <i>Can Language Models Ask Useful Questions?</i>	July 2025
University of Washington ZLab, <i>Eliciting Tasks and Preferences with Language Models</i>	August 2024

Mentorship and Teaching

Research Mentorship

1. Itamar Pres (MIT PhD)	September 2025 –
2. Amiri Hayes (MIT Summer Research Program)	June 2025 –
3. Carl Guo (MIT UROP; MIT MEng)	Sep. 2024 –
4. Nolan Clement (MIT UROP)	Sep. 2024 – May 2025
5. Maxine Liu (MIT Summer Research Program; now PhD @ UPenn)	June 2024 – Feb. 2025
6. Kunal Handa (Brown BS; Oxford Masters; now @ Anthropic)	July 2023 – March 2024
7. Will Chen (MIT MEng; now PhD @ UC Berkeley)	May 2022 – Feb. 2023
8. Nitya Parthasarathy (MIT UROP)	Sep. 2020 – June 2021

Teaching Assistant

Massachusetts Institute of Technology

1. 6.861*: Quantitative Methods for NLP	Fall 2023
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University of Washington

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|---|-------------|
| 1. CSE 311: Foundations of Computing I (Discrete Math), with Kevin Zatloukal | Spring 2019 |
| 2. CSE 312: Foundations of Computing II (Probability/Statistics), with Martin Tompa | Winter 2019 |
| 3. CSE 312: Foundations of Computing II (Probability/Statistics), with Martin Tompa | Winter 2018 |
| 4. CSE 331: Software Design and Implementation, with Kevin Zatloukal | Autumn 2017 |

Service

Workshop Organizing

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| - Assessing World Models: Methods and Metrics for Evaluating Understanding, <i>Co-lead Organizer</i> | ICML 2025 |
| - Wordplay: When Language Meets Games, <i>Co-organizer</i> | ACL 2024 |

Program Committee – Reviewer

- ICLR 2024, 2025
- ICML 2024, 2025
- ACL 2023
- ARR February 2025; December, October, June, April, February 2024
- COLING 2022, 2020
- EMNLP 2022, 2021, 2020

MIT Summer Research Program – Mentor

2024, 2025

- Provided research mentorship for undergrads in under-represented and marginalized groups from across the US

MIT EECS Department Graduate Admissions Committee

2023

- Reviewed graduate school applications to the MIT EECS department, Artificial Intelligence & Decision-making research area

MIT GAAP Mentor

2020, 2021

- Mentoring individuals from underrepresented groups through the graduate application process.

Student Volunteer/Scholarship

- ACL 2021, 2020

Last Updated

November 24, 2025